

CE 310

Shape Matters — Reading Distributions

Week 2 · Histograms · Skewness · Box Plots · Percentiles

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Part 1

Why Shape Matters

A Grid Failed — 246 People Died

February 2021 · Texas Winter Storm Uri

- Temperatures dropped to **-2°F** in parts of Texas
- **40+ power plants** went offline within 48 hours
- 246 deaths · \$195 billion in damages



*NOAA satellite: Houston goes dark, Feb 16, 2021 —
NOAA/NESDIS, public domain.*

The engineers had the data — every past cold snap was in the historical record. **The failure:** sizing the grid for **average winter temperature**, ignoring a dangerous **left tail**.

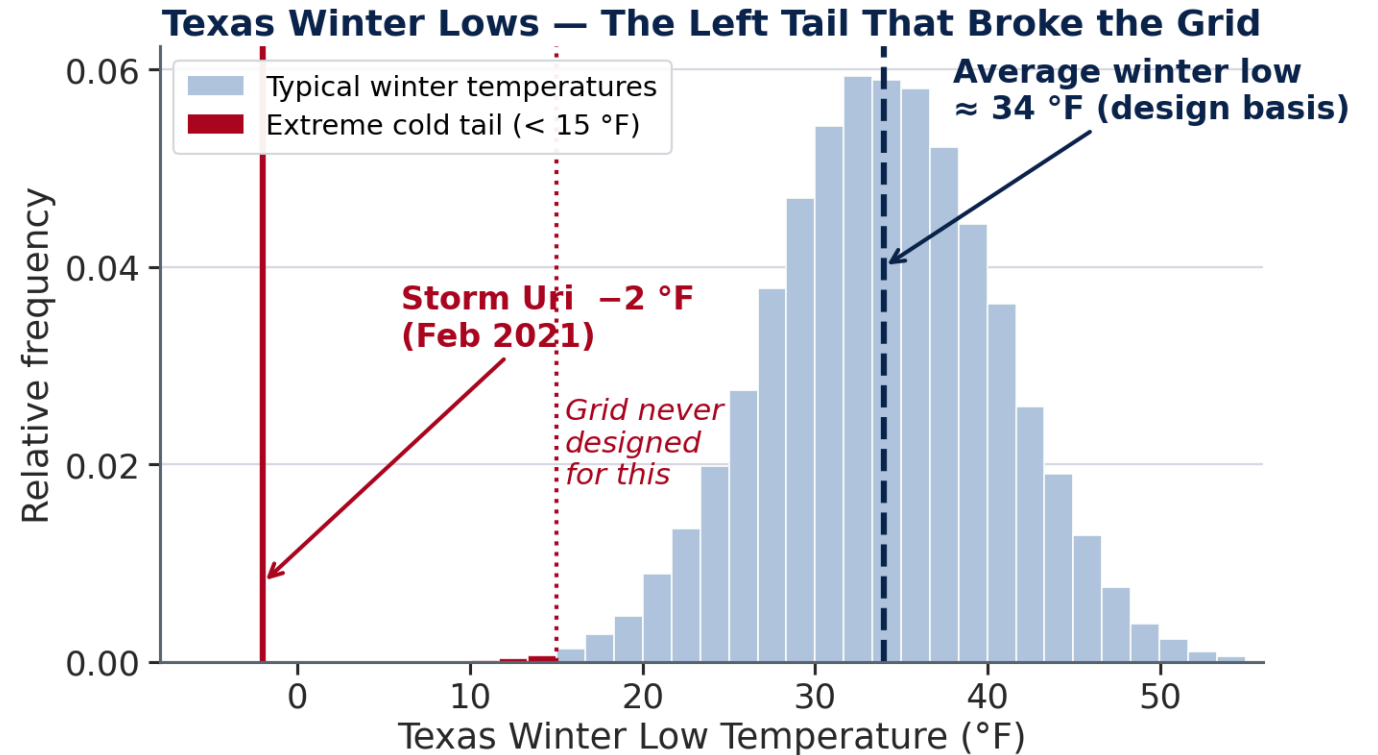
! The lesson: The average tells you where the middle is. It tells you nothing about how bad the extremes can get — or how often.

The Left Tail They Ignored

The distribution of Texas winter temperatures has a **long left tail** — rare but extreme cold events.

- **Average** winter temperature: mild, manageable
- **Left tail** events: occur once per decade — but catastrophic when they do

Using the average as the design basis means you are **never prepared for the tail**.



Your Turn — Predict the Shape

Before we look at the data: based on what you know about Tucson's climate, predict the shape of each distribution.

Variable	Your prediction: Left / Symmetric / Right?
Annual temperature (hourly, full year)	???
Wind speed (hourly, full year)	???
Relative humidity (hourly, full year)	???

Think about it for 90 seconds. We'll reveal the answer after the histogram slides.

✓ Predict the Shape — Answers

Tucson 2023 hourly data

Variable	Actual shape	Physical reason
Temperature	Nearly symmetric	Cold winter left tail $\approx$ hot summer right tail
Wind speed	Right-skewed	Speed cannot be negative; calm hours dominate, rare storms pull right
Relative humidity	Right-skewed	Desert is mostly dry; monsoon season pulls the right tail

 **Right-skewed** (positive skew): the distribution has a long tail on the right side. A few unusually high values pull the mean above the median. Most observations cluster at lower values; rare high extremes are possible.

Three Design Decisions — All Require Knowing Shape

Design Decision	Wrong approach	What you actually need
Size HVAC cooling system	Annual average temp (72°F)	95th percentile temp (100°F)
Design wind load for signage	Average wind speed (7.2 mph)	95th percentile wind (17.2 mph)
Size dehumidification coil	Annual average RH (32.7%)	90th percentile RH (ASHRAE 62.1)
Plan heating capacity (TX)	Assume symmetric winter temps	Recognize the left tail

💡 **Pattern:** Every design-for-worst-case problem requires knowing the tail — not the center. The center only tells you what is typical.

Part 2

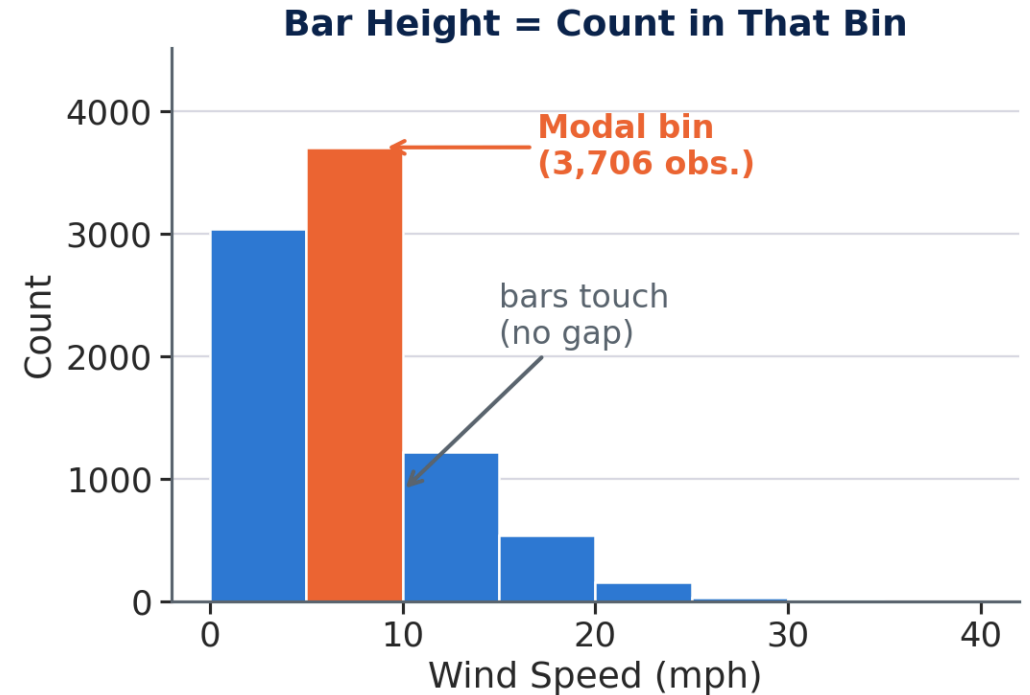
The Histogram

What Is a Histogram?

 **Definition:** A histogram divides a continuous variable into equal-width **bins** and counts how many observations fall in each bin. The height of each bar = count of observations in that bin.

Key properties:

- Bins are **equal width**; bars **touch** — no gap (unlike a bar chart)
- x-axis: **continuous scale**; y-axis: **count** or frequency
- Reveals **shape**, **clusters**, and **outliers**



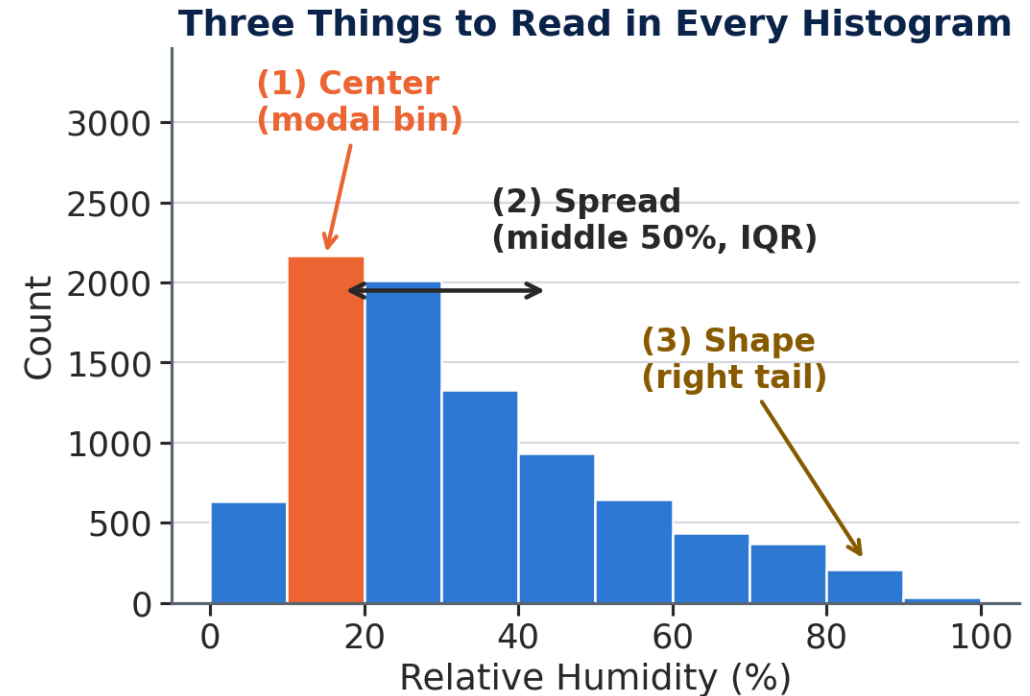
Reading a Histogram — What to Look For

(1) Center — where does the bulk of the data sit? Is the peak near the mean, or off to one side?

(2) Spread — how wide is the distribution? Values clustered tightly or spread over a wide range?

(3) Shape — symmetric? One tail longer than the other? One peak or two?

⚠ A symmetric distribution and a right-skewed one can share the same mean and SD — shape reveals what those numbers hide.

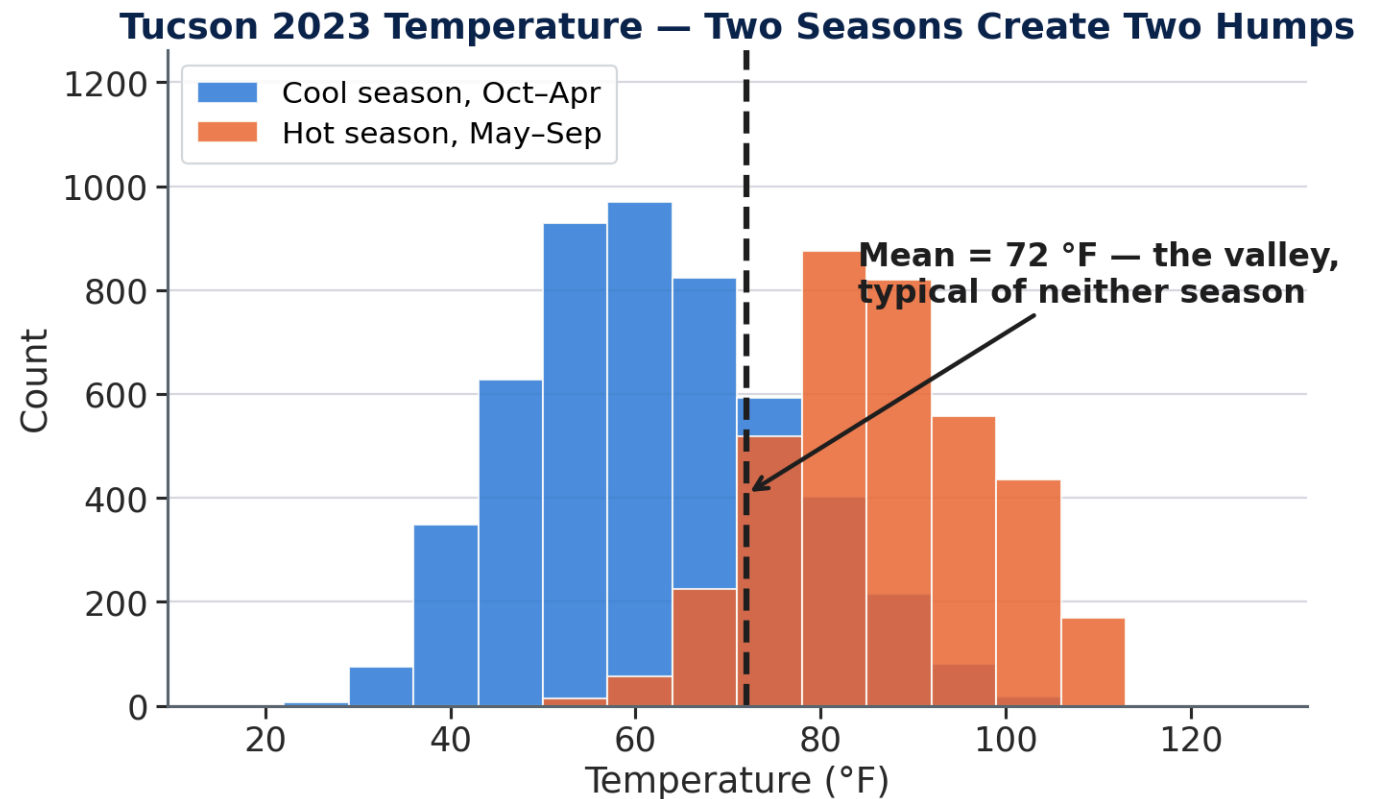


One Peak or Two? — Bimodal Distributions

Most distributions have **one peak** (unimodal). But mixing two subpopulations creates **two peaks** (bimodal):

Common causes:

- Two distinct seasons or occupancy modes
- Equipment with two operating states (on vs. standby)
- A process that changed mid-measurement



One Peak or Two? — Reading the Warning Signs

Key warning signs of a bimodal distribution:

- Two humps with a valley between them
- Mean falls **in the valley** — typical of neither group
- $SKEW \approx 0$ can mislead: a symmetric bimodal still has two distinct typical values

⚠ Two humps signal a mixed population — split by subgroup before computing any statistics.

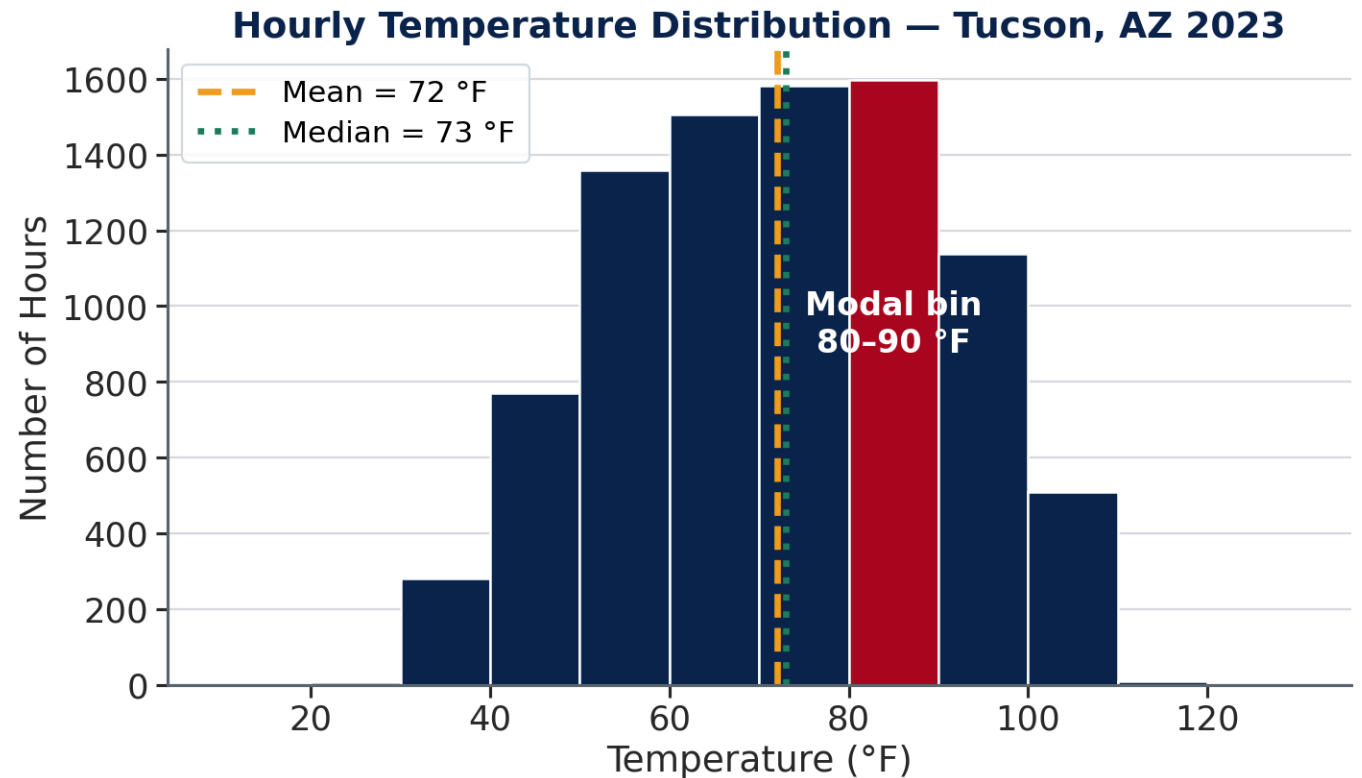
Tucson Temperature — What Does the Histogram Show?

- **Modal bin: 80–90°F** — most common temperature range
- **Mean (72°F) ≠ Mode (85°F)** — they differ by 13°F
- **Plateau shape**, not a bell curve

Why the plateau?

Two overlapping seasons:

- Cool season (Dec–Mar): 40–65°F
- Hot season (Jun–Sep): 85–110°F



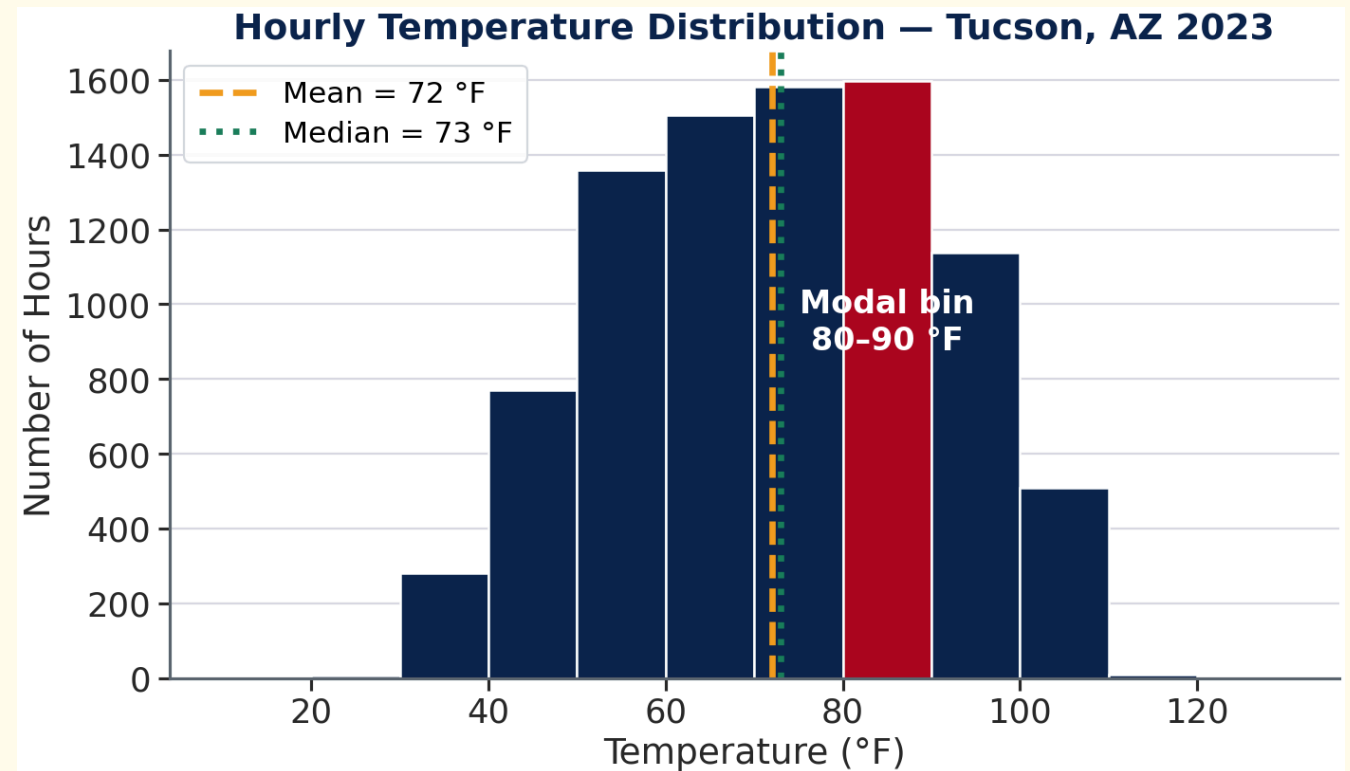
Look at This Histogram

Based on what you see — is this distribution:

A) Left-skewed B) Symmetric C) Right-skewed

Clues to check:

- Where is the **peak** (tallest bar)?
- Is one **tail longer** than the other?
- Is the mean (**gold dashed**) to the left or right of the median (**white dotted**)?



✓ Answer — Nearly Symmetric (SKEW = -0.05)

The mean (72°F) and median (73°F) differ by only 1°F — a near-perfect match, the hallmark of symmetry.

Why it looks left-skewed but isn't:

- Modal bin ($80\text{--}90^{\circ}\text{F}$) is above the mean
- This looks like left skew — but the **cold winter tail** ($20\text{--}50^{\circ}\text{F}$) nearly cancels the **hot summer tail** ($90\text{--}110^{\circ}\text{F}$)
- Net result: almost symmetric

What would be surprising about wind?

- Wind speed **cannot be negative**
- Most hours: calm ($0\text{--}10$ mph)
- Occasional storms: $20\text{--}34$ mph
- This forces a **right tail** → right-skewed

💡 A SKEW coefficient near zero (-0.05) confirms what the mean $\approx$ median already suggested.

Part 3

Measuring Shape — Skewness

What Skewness Measures

 **Skewness** quantifies the asymmetry of a distribution. It tells you whether one tail is longer than the other — and by how much.

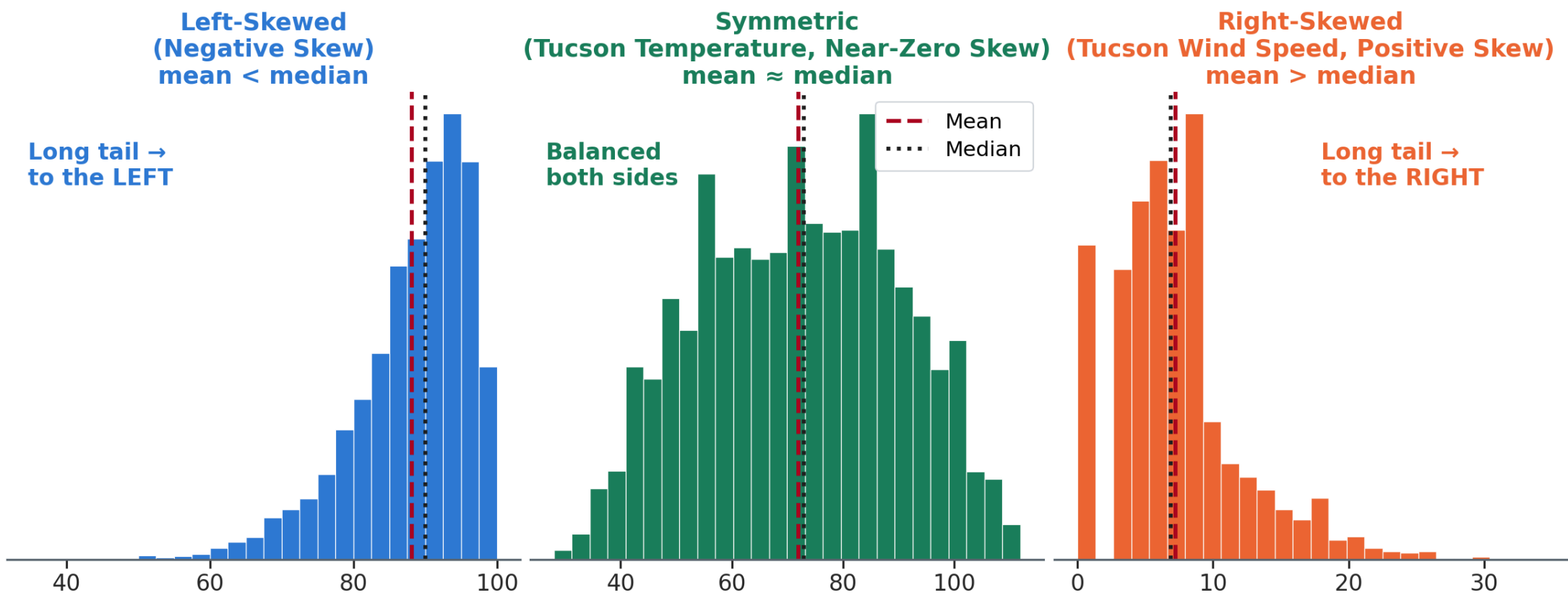
$$g = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

x_i = each value · $\bar{x}$ = sample mean · s = sample standard deviation

Interpretation guide:

SKEW value	Meaning
Near 0 (−0.5 to +0.5)	Symmetric — mean $\approx$ median
Negative (< −0.5)	Left-skewed — long left tail, mean < median
Positive (> +0.5)	Right-skewed — long right tail, mean > median

The Three Shapes



Left Skew vs. Right Skew — What Pulls the Mean

Left-skewed (negative)

- Tail extends to the LEFT
- A few very LOW values pull the mean DOWN
- Mean < Median
- Example: exam scores where most students do well but a few fail badly

Engineering example: Strength of materials — most samples are at or above spec, but rare defects create a low-value left tail

Right-skewed (positive)

- Tail extends to the RIGHT
- A few very HIGH values pull the mean UP
- Mean > Median
- Example: income — most people earn modest amounts, a few billionaires pull the average up

Engineering example: Wind speed, rainfall, flood flows — most hours are calm, but rare storms create a high-value right tail

The Three Tucson Variables — Revealed

Variable	Mean	Median	SKEW	Shape
Temperature (°F)	72	73	−0.05	Nearly symmetric
Wind speed (mph)	7.2	6.9	+1.07	Right-skewed
Relative humidity (%)	32.7	27.6	+0.92	Right-skewed

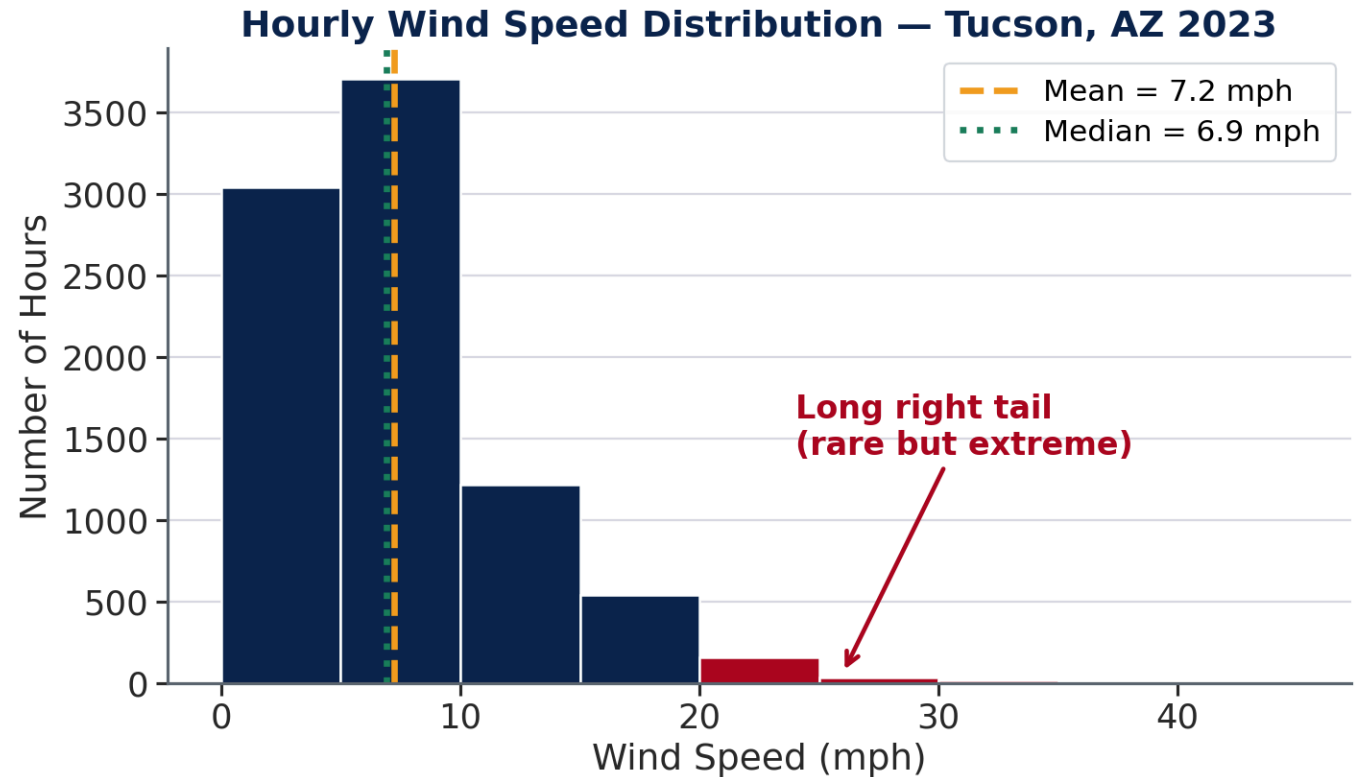
💡 **Wind and humidity are both right-skewed** — most hours are calm/dry, but rare monsoon and storm events create the high-value right tails. This is why using the *average* wind or humidity for design grossly underestimates the worst conditions.

Wind Speed — The Right Skew in Action

Wind speed statistics:

- Mean: 7.2 mph · Median: 6.9 mph
- SKEW: **+1.07** (right-skewed)
- P95: **17.2 mph** (2.4× the mean)
- **P95 wind pressure = 5.7× average** (wind pressure $\propto V^2$)

Why right-skewed? Speed cannot be negative — bounded at zero. Most hours are calm; rare storms form the right tail.



Quick Check — Classify the SKEW

Three datasets, three SKEW values. Match each SKEW to the variable — and explain the physical reason for that shape.

Dataset	SKEW	Your call: Left / Symmetric / Right?
Dataset X	-1.4	???
Dataset Y	+0.1	???
Dataset Z	+2.3	???

Which variable is which?

- **A)** Concrete cylinder compressive strength (100 lab-controlled samples)
- **B)** Monthly salary at a 500-person engineering firm
- **C)** Daily peak water demand at a city treatment plant

✓ Quick Check — Answers

Dataset	SKEW	Variable	Physical reason
X	-1.4	Concrete strength	Most samples meet or exceed spec; rare defective batches pull the left tail down
Y	+0.1	Monthly salary	Salaries in a single firm cluster near the median; few extreme outliers → near symmetric
Z	+2.3	Daily water demand	Most days near baseline; rare heat waves or large events create a long right tail

💡 **Physical reasoning predicts shape before you plot.** Ask: Is there a zero floor? (→ right skew.) Is there a quality target most samples exceed? (→ left skew.) Is the process tightly controlled? (→ symmetric.)

Part 4

The Box Plot

The Five-Number Summary

In Week 1 you computed Q1, Q3, and IQR for Tucson temperatures. Those five numbers — Minimum, Q1, Median, Q3, Maximum — are the **five-number summary**: the foundation of every box plot.

For Tucson 2023 temperatures:

Statistic	Value	Meaning
Minimum	28°F	Coldest single hour of the year
Q1	58°F	25% of hours are colder than this
Median	73°F	Half of hours are above, half below

The Five-Number Summary – Upper Half

For Tucson 2023 temperatures (continued):

Statistic	Value	Meaning
Q3	86°F	75% of hours are colder than this
Maximum	112°F	Hottest single hour of the year
IQR	28°F	Spread of the middle 50% — Q3 – Q1

💡 These five numbers — Min, Q1, Median, Q3, Max — map directly onto the box plot: the box spans Q1 to Q3, the line inside is the median, and the whiskers extend toward the Min and Max.

Reading a Box Plot

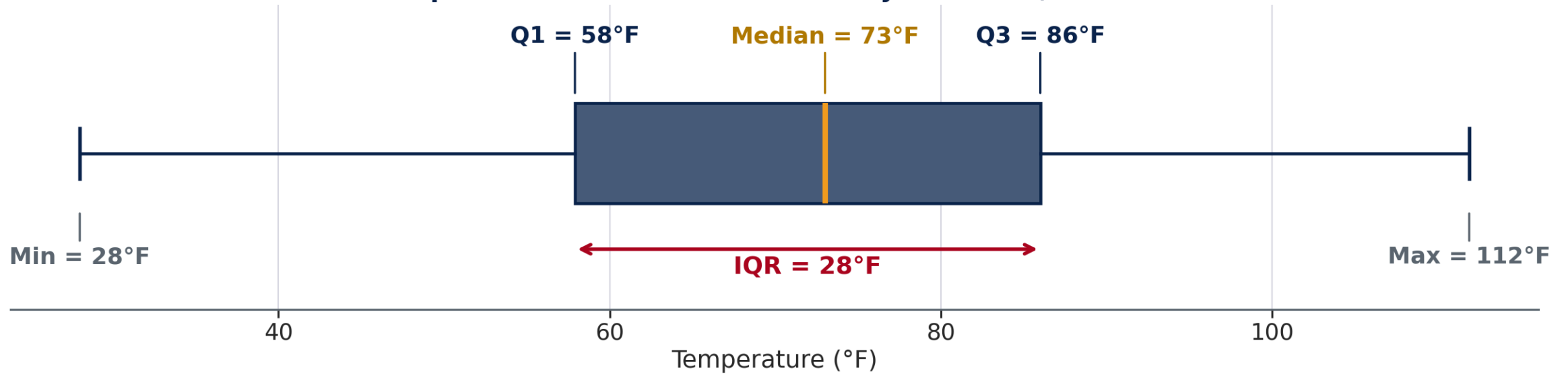
The box spans Q1 to Q3 — the middle 50% (IQR)

The whiskers extend to the most extreme values within $1.5 \times \text{IQR}$

The line inside is the median (not the mean)

Dots beyond whiskers are outliers

Temperature Five-Number Summary — Tucson, AZ 2023



IQR vs. Standard Deviation — What's the Difference?

Standard Deviation (SD)

- Measures spread of **all** values
- Heavily influenced by **outliers** and **extreme values**
- Assumes roughly symmetric distribution for easy interpretation
- Tucson temperature SD: **17.9°F**

Interquartile Range (IQR)

- Measures spread of the **middle 50%**
- **Outlier-resistant** — extreme values don't affect it
- Works well for skewed distributions
- Tucson temperature IQR: **28°F**

💡 For right-skewed data (wind, humidity, rainfall), IQR is a more reliable spread measure than SD because it ignores the long tail. For symmetric data (temperature here), SD and IQR give complementary information.

Quick Poll — IQR vs. SD

Two cities have the same annual average temperature (72°F) and the same standard deviation (18°F). But City A has an IQR of 28°F and City B has an IQR of 12°F .

What does this tell you about the two cities?

- A) They have identical climates — same mean and SD means same distribution
- B) City A has a more uniform climate across the year; City B has more concentration near the average
- C) City B has a more concentrated middle 50% — tighter "typical" temperature band; City A has wider seasonal swings
- D) Not enough information to say anything

✓ Quick Poll — The Answer Is C

City B has the tighter typical temperature band

	City A	City B
Annual mean	72°F	72°F
Standard deviation	18°F	18°F
IQR	28°F	12°F
Climate character	Wide seasonal swings	Narrow "typical" band

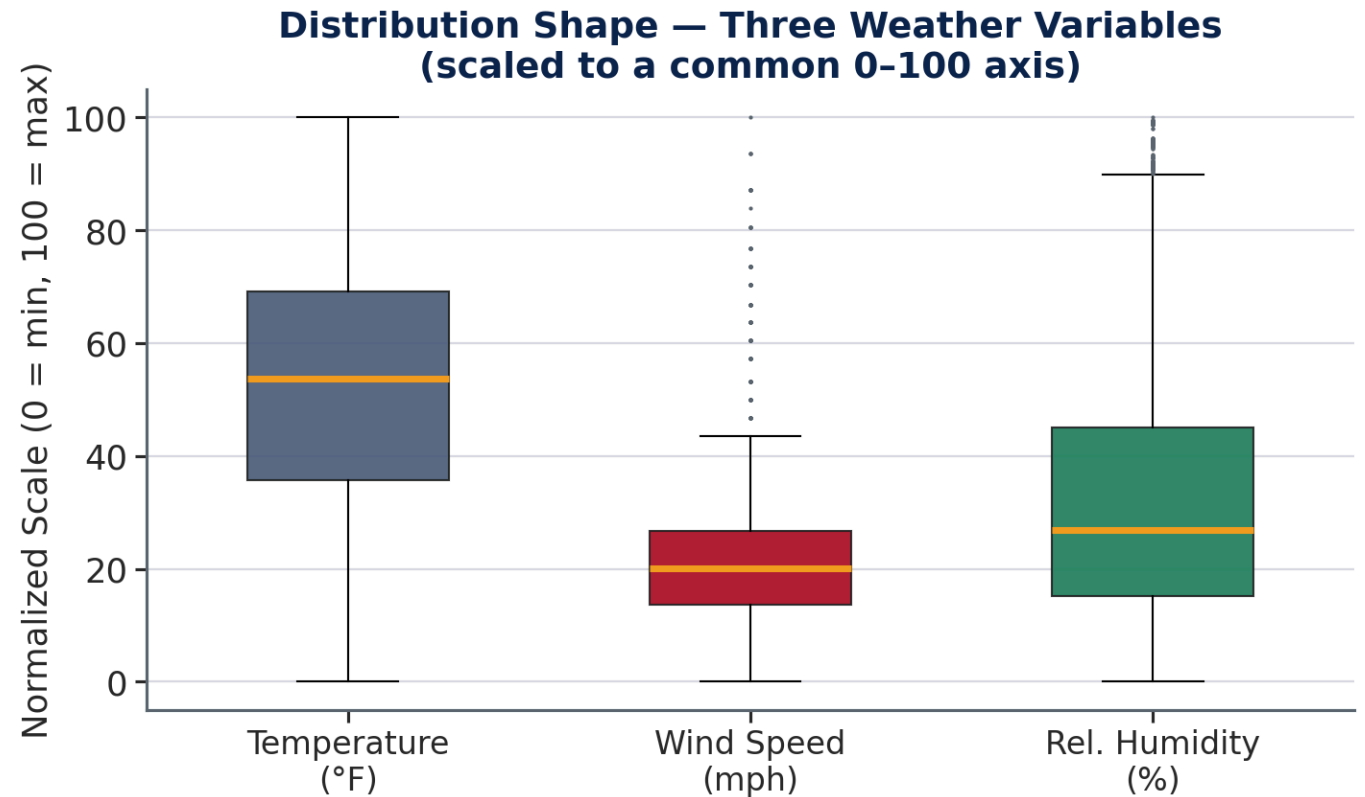
Why same SD but different IQR? City A's large IQR reflects broad seasonal swings (cold January, hot July); City B's hours cluster near the mean, offset by occasional extremes that push the SD up.

💡 SD reacts to extremes; IQR reflects the middle 50%.

Box Plots Side by Side — Three Variables

Three variables, all normalized scale:

- Temperature: **wide, symmetric box** — large seasonal range, IQR = 28°F
- Wind speed: **narrow box, long upper whisker** — most hours calm, IQR = 4.5 mph
- Humidity: **narrow box, long upper whisker** — IQR $\approx$ 34%



Part 5

Percentiles and Design Thresholds

Design Percentiles — The Full Spectrum

In Week 1 you found the ASHRAE 1% design temperature — the 88th-largest of 8,760 values, the 99th percentile by rank. Here is what the full spectrum looks like for Tucson:

Tucson 2023:

Percentile	Temperature	Meaning
25th (Q1)	58°F	25% of hours are below 58°F
50th (Median)	73°F	Half the year is below 73°F
75th (Q3)	86°F	75% of hours are below 86°F
90th	96°F	Only 10% of hours exceed 96°F
95th	100°F	HVAC cooling design threshold (5% exceedance)

Design Percentiles — Why Not 100%?

Engineering design is always a tradeoff between **cost** and **risk**:

100th percentile (absolute maximum)

- Guarantees performance under every observed condition
- Extremely expensive to design for
- The maximum may be a freak event never to repeat

90th–99th percentile

- Handles the vast majority of conditions
- Much more economical
- Residual risk is **known and accepted**

Standard engineering practice:

- HVAC cooling loads: **99th or 97.5th** percentile dry-bulb (ASHRAE)
- Structural wind loads: **50-year return period** (~98th annual percentile)

Precision Check — Percentile ≠ Probability

 **P90 does not mean "90% probability."** It means 90% of observed values fall *below* this point — a rank in your data, not a forecast.

Formal statement: The Pth percentile is the value below which P% of all observations fall; (100 – P)% are above it.

Statement	Plain English
P90 of wind speed = 14.8 mph	90% of all hours are calmer than 14.8 mph
P95 of temperature = 100°F	5% of all hours are hotter than 100°F
P99 of temperature = 107°F	1% of all hours exceed 107°F — ASHRAE cooling design threshold

The key insight: A percentile is a **location** in the distribution — not a probability, not a maximum.

From Definition to Number — The Logic Behind a Percentile

Example: Find the 90th percentile of Tucson wind speed (8,760 hourly values)

Step 1 — How many values fall below P90?

$$8,760 \times 0.90 = 7,884 \text{ values}$$

Step 2 — What is the 7,884th-lowest value?

Sort low to high → position 7,884 → **P90 ≈ 14.8 mph**

Step 3 — Verify the interpretation:

7,884 out of 8,760 hours have wind speed ≤ 14.8 mph ✓

876 hours (10%) have wind speed > 14.8 mph ✓

💡 This is the arithmetic the computer carries out when you ask it for a percentile. The tool does the sorting and counting — you supply the judgment about *which* percentile the design requires.

Exceedance Probability — The Language of Design Standards

Exceedance probability = the fraction of time a threshold is exceeded.

$$P(\text{exceedance}) = 1 - \frac{p}{100} \quad (p = \text{design percentile, e.g. } p = 95 \Rightarrow 5\% \text{ exceedance})$$

Design percentile	Wind speed	Exceedance	Hours/year above threshold
P90	14.8 mph	10%	876 hr/yr
P95	17.2 mph	5%	438 hr/yr
P99	21.4 mph	1%	88 hr/yr

Why engineers use this language:

- ASHRAE: "0.4% cooling condition" = P99.6 temperature

💡 Design standards are written in exceedance language — translate any percentile to hours per year.

Quick Check — Design Threshold Decision

A Tucson building engineer sized a dehumidification coil to handle humidity at the **90th percentile (RH = 58.4%)**.

Two questions:

(a) The dataset has 8,760 hours. How many hours per year will outdoor humidity **exceed** the coil's design capacity?

(b) The building owner asks: "*Will the dehumidification system always keep us comfortable?*" How do you answer — honestly and professionally?

✓ Design Threshold — Answers

(a) Hours above design capacity per year:

$10\% \times 8,760 = 876 \text{ hours/year}$ (≈ 36 days — primarily July–August monsoon season)

(b) The honest professional answer:

"Not always. The system is designed to the ASHRAE 62.1 90th-percentile standard, meaning outdoor humidity will exceed the coil's capacity roughly 876 hours per year. This is the industry-accepted level of risk. A system designed for the 99th percentile would handle 99% of hours but would cost substantially more to build and operate. The choice of design percentile is a cost-risk tradeoff made explicitly at design time."

The principle:

Every design percentile is a **decision to accept a known exceedance frequency**.

Part 5b

Putting It All Together

A Complete Design Workflow — Four Tools, One Decision

Scenario: A civil engineer must specify the design wind speed for a Tucson billboard structural attachment. City code requires the **95th percentile** wind speed as the design basis.

The four-step workflow — executed in order:

Step	Tool	Question answered
1	Histogram	What is the shape of the wind distribution?
2	SKEW coefficient	Which direction — and by how much?
3	Box plot	Where is the middle 50%? Are extreme values real?
4	Percentile	What is the actual P95 value for the design basis?

Today's lab (Sections B7–B11) puts you through exactly this sequence.

Steps 1 & 2 — Shape: Histogram and SKEW

What the wind histogram reveals:

Histogram reading:

- **Modal bin: 0–5 mph** — most hours are calm
- **Long right tail** extending beyond 20 mph
- Very few hours above 25 mph — but they exist
- Hard floor at zero: speed cannot be negative

Step 2 — Confirm with SKEW:

$$\text{SKEW} = +1.07$$

- ✓ Positive → right-skewed
- ✓ Magnitude > 0.5 → meaningful asymmetry

The mean (7.2 mph) sits far below the dangerous tail — designing for average speed ignores the structurally critical hours.

⚠ Right-skewed data: the mean underestimates the design condition. You must go to the tail.

Steps 3 & 4 — Box Plot and Percentile

Step 3 — What the box plot confirms:

Box plot feature	Value	Engineering meaning
IQR (Q1 to Q3)	4.5 mph	Middle 50% of hours span only 4.5 mph — most time is calm
Upper whisker endpoint	~17 mph	Typical non-extreme upper bound
Outlier dots	> 17 mph	Verified extreme events — real, not sensor errors

Step 4 — Find the 95th percentile:

$8,760 \times 0.95 = 8,322$ nd-lowest value → **P95 = 17.2 mph** (438 hr/yr above this)

💡 Wind pressure $\propto V^2$. At P95 (17.2 mph): $17.2^2 = 296$ units. At the mean (7.2 mph): $7.2^2 = 52$ units.
Ratio: 5.7× — designing for average speed produces a structure 5.7× underbuilt at P95 conditions.

The Workflow — Why Order Matters

Four tools, used in sequence, answered one engineering question:

Step	Finding	Decision it enables
Histogram	Right-skewed, long upper tail	Cannot use the mean as design basis
SKEW = +1.07	Meaningfully asymmetric	A percentile is required — not a summary statistic
Box plot	IQR = 4.5 mph; outliers confirmed real	Extreme events are not measurement errors — include them
P95 = 17.2 mph	438 hr/yr above threshold	Design basis confirmed; proceed to structural calculation

💡 The sequence matters. Skipping to step 4 without steps 1–3 means you know P95 = 17.2 mph but cannot explain *why* the mean fails.

The Texas grid engineers knew the average. What they skipped: step 1.

What Is Exploratory Data Analysis (EDA)?

 **EDA** is the practice of examining a dataset — through summaries and visualizations — before drawing conclusions or building models. Coined by statistician John Tukey (1977).

The four goals of EDA:

Goal	Question you're asking
Describe	What are the center, spread, and shape of each variable?
Discover	Are there patterns, clusters, or seasonal structure?
Check	Do the data meet the assumptions of the analysis I plan to run?
Flag	Are there outliers, gaps, or errors that could distort results?

EDA is not a formal test — it is a conversation with your data before you commit to an analysis. Today's lab puts all four goals into practice: **Describe** (A1–A8 · summary statistics), **Discover** (histogram patterns), **Check** (skewness, box plots), **Flag** (outlier dots).

Part 6

Today's Lab

What You'll Do Today — Section A

Dataset: Same Tucson 2023 file from Week 1

Section A — Histogram Basics

- A1–A2: Count hours in specific temperature bins
- A3: Verify your bin counts with a frequency array
- A4: Measure temperature skewness
- A5–A6: Wind speed bin counts
- A7: Wind speed skewness
- A8: Humidity skewness

What You'll Do Today — Section B

Section B — In-Depth Analysis

- B1–B3: Temperature Q1, Q3, IQR
- B4–B5: Temperature 90th and 95th percentile
- B6: Hours above the 95th percentile
- B7–B11: Wind quartiles, IQR, P95, wind pressure at P95
- B12–B13: Humidity P90, August average humidity
- B14: Probability bridge — estimate $P(80 \leq T < 90)$ from your histogram

In-Lab Target — What to Finish Before Class Ends

Take-home (before deadline — see D2L):

- B7–B14: wind analysis, humidity percentiles, probability bridge
- Chart 2: Box-and-whisker plot (Excel 2016+ built-in)
- Memo PDF: 3 paragraphs on distribution shape and design implications

💡 B14 — Probability Bridge: $P(80 \leq T < 90)$ from your histogram = relative frequency. Week 3 formalizes this.

Looking Ahead — Week 3 Preview

At the end of today's lab you'll see this question (ungraded):

"In last week's lab you counted the hours where outdoor temperature exceeded 95°F. Use that count to estimate $P(T > 95^\circ\text{F})$."

$$P(T > 95^\circ\text{F}) = 900 \text{ hours} / 8,760 \text{ hours} = \mathbf{10.3\%}$$

The question to sit with over the week:

- Is this the **true probability** of a hot hour in Tucson?
- Or is it an **estimate** based on one year of data?
- What would you need to be more confident?

💡 Week 3 is about conditional probability — $P(A | B)$ — and will build directly on the histograms and percentiles you compute today.

Summary — Week 2

Shape is information that summary statistics discard

The mean and standard deviation describe center and spread — but two distributions with identical mean and SD can look completely different when you plot them.

Today's tools:

- **Histogram** — visualize the full distribution
- **SKEW coefficient** — quantify asymmetry
- **Box plot** — five-number summary at a glance
- **Percentile** — locate any point in the distribution

The core principle: Design for the **tail**, not the center. The histogram and percentile are how you find the tail.